AMCS Practice 4

Sums 2. Summation factors. Efficiency of quicksort. Manipulation of sums. Multiple- sums.

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Summation Factors

Formula 1. General Linear Recurrence

A linear recurrence can be written in the form

$$\boxed{a_n T_n = b_n T_{n-1} + c_n}$$

where:

- T_n — the value of the sequence that we want to find;
- T_{n-1} — the previous value of the sequence;
- a_n — the coefficient of T_n;
- b_n — the coefficient of T_{n-1};
- c_n — an additional term;
- T_0 — the initial value.

This is the general form used for the **summation factor method**.

How to use it

First, rewrite the recurrence in the form

$$a_n T_n = b_n T_{n-1} + c_n$$

and identify a_n, b_n, and c_n.

Example 1

For the recurrence

$$T_n = 3T_{n-1} + 2,$$

identify a_n, b_n, c_n.

Example 2

For

$$2T_n = 5T_{n-1} + n,$$

identify a_n, b_n, c_n.

Example 3

For

$$(n+1)T_n = (n+2)T_{n-1} + 3n,$$

identify a_n, b_n, c_n.

Summation Factor Condition

The main formula is

$$\boxed{s_n b_n = s_{n-1} a_{n-1}}$$

Therefore,

$$\boxed{s_n = \frac{s_{n-1} a_{n-1}}{b_n}}$$

provided that $b_n \neq 0$.

What do the symbols mean?

- s_n — the summation factor at step n;
- s_{n-1} — the previous summation factor;
- a_{n-1} — the value of a_n at $n-1$;
- b_n — the coefficient of T_{n-1}.

We usually start with

$$\boxed{s_0 = 1}.$$

How to use it

For

$$a_n T_n = b_n T_{n-1} + c_n :$$

1. Identify a_n and b_n.
2. Set $s_0 = 1$.
3. Calculate $s_1, s_2, s_3, \ldots$.

Example 1

Given

$$T_n = 2T_{n-1} + 1,$$

find s_1, s_2, s_3.

Example 3

Given

$$(n+1)T_n = (n+2)T_{n-1} + n,$$

find s_1, s_2, s_3.

General Formula for the Summation Factor

Instead of calculating each s_n separately, we can use

$$\boxed{s_n = \frac{a_0 a_1 \cdots a_{n-1}}{b_1 b_2 \cdots b_n}}$$

when $s_0 = 1$.

How to use it

Substitute the values of a_k and b_k into the products and simplify.

For example, if

$$a_n = 1, \qquad b_n = 2,$$

then

$$s_n = \frac{1 \cdot 1 \cdots 1}{2 \cdot 2 \cdots 2}.$$

Example 1

Given

$$a_n = 1, \qquad b_n = 4,$$

find s_n.

Example 2

Given

$$a_n = 2, \qquad b_n = 5,$$

find s_n.

Example 3

Given

$$a_n = n + 1, \qquad b_n = n + 2,$$

find and simplify s_n.

Closed Formula for a Recurrence

After finding the summation factor, we can use

$$\boxed{T_n = \frac{1}{s_n a_n}\left(s_0 a_0 T_0 + \sum_{k=1}^{n} s_k c_k\right)}$$

What do the symbols mean?

- T_n — the value we want to find;
- T_0 — the initial value;
- s_n — the summation factor;
- a_n — the coefficient of T_n;
- c_k — the additional term;
- $\sum s_k c_k$ — the accumulated contribution of the additional terms.

How to use it

The general procedure is

$$\boxed{\text{Recurrence} \to a_n, b_n, c_n \to s_n \to T_n}$$

Example 1

Solve

$$T_n = 2T_{n-1} + 1, \qquad T_0 = 0.$$

Example 2

Solve

$$T_n = 3T_{n-1} + 2, \qquad T_0 = 1.$$

Example 3

Solve

$$T_n = 2T_{n-1} + n, \qquad T_0 = 0.$$

Constant-Coefficient Recurrence

An important special case from the lecture is

$$\boxed{Z_n = aZ_{n-1} + b}.$$

Its solution is

$$\boxed{Z_n = a^n Z_0 + b\frac{a^n - 1}{a - 1}} \qquad (a \neq 1).$$

What do the symbols mean?

- Z_n — the value at step n;
- Z_0 — the initial value;
- a — the constant multiplier;
- b — the constant added at each step.

How to use it

Identify a, b, and Z_0, then substitute them directly into the formula.

Example 1

$$Z_n = 2Z_{n-1} + 1, \qquad Z_0 = 0.$$

Find Z_n.

Example 2

$$Z_n = 3Z_{n-1} + 4, \qquad Z_0 = 2.$$

Find Z_n.

Example 3

$$Z_n = 5Z_{n-1} - 2, \qquad Z_0 = 3.$$

Find Z_n.

Efficiency of Quicksort

The lecture gives the average number of comparisons in Quicksort as

$$\boxed{C_n = n + 1 + \frac{2}{n}\sum_{k=0}^{n-1} C_k, \qquad C_0 = 0}$$

where C_n is the **average number of comparisons for n elements.**

Main Idea

When a recurrence contains a sum such as

$$\sum C_k,$$

we can:

1. Write the equation for n.
2. Write the corresponding equation for $n-1$.
3. Subtract the equations.
4. Eliminate the sum.

For Quicksort, this gives

$$\boxed{nC_n = (n+1)C_{n-1} + 2n}$$

Example 1

Using

$$C_n = n + 1 + \frac{2}{n}\sum_{k=0}^{n-1} C_k, \qquad C_0 = 0,$$

find C_1.

Example 2

Using the same recurrence, calculate

$$C_1, \quad C_2, \quad C_3.$$

Example 3

Starting from

$$C_n = n + 1 + \frac{2}{n}\sum_{k=0}^{n-1} C_k,$$

derive

$$nC_n = (n+1)C_{n-1} + 2n.$$

Harmonic Numbers and Quicksort

For Quicksort, the lecture obtains

$$\boxed{C_n = 2(n+1)H_n - 2n}$$

where

$$\boxed{H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \sum_{k=1}^{n} \frac{1}{k}}$$

is the n-th **harmonic number**.

How to use it

First calculate H_n. Then substitute it into the formula for C_n.

Example 1

Find

H_3.

Example 2

Use

$$C_n = 2(n+1)H_n - 2n$$

to find C_4.

Example 3

Calculate and compare C_5 and C_{10} using the closed formula.

Manipulation of Sums — Constant Factor

The first rule is

$$\boxed{\sum_{k\in K} ca_k = c\sum_{k\in K} a_k}$$

A constant factor can be taken outside the summation sign.

How to use it

If every term in a sum contains the same constant multiplier, move that constant outside the sum.

Example 1

Simplify:

$$\sum_{k=1}^{10} 3k.$$

Example 2

Evaluate:

$$\sum_{k=1}^{5} 4k^2.$$

Example 3

Evaluate:

$$\sum_{k=1}^{6} (5k+2).$$

Splitting a Sum

The second rule is

$$\sum_{k\in K}(a_k + b_k) = \sum_{k\in K} a_k + \sum_{k\in K} b_k$$

How to use it

If the expression inside the summation contains several terms, we can split it into separate sums.

Example 1

Evaluate:

$$\sum_{k=1}^{5}(k+2).$$

Example 2

Evaluate:

$$\sum_{k=1}^{6}(k^2+3k).$$

Example 3

Evaluate:

$$\sum_{k=1}^{10}(2k^2-3k+4).$$

Changing the Index

The order of the terms in a finite sum can be changed:

$$\boxed{\sum_{k\in K} a_k = \sum_{p(k)\in K} a_{p(k)}}$$

where $p(k)$ represents a permutation of the indices.

A particularly useful substitution is

$$k \to n - k.$$

How to use it

For example,

$$\sum_{k=0}^{n} f(k)$$

can be rewritten as

$$\sum_{k=0}^{n} f(n-k).$$

This is useful when the reversed form can be combined with the original sum.

Example 1

Rewrite using $j = n - k$:

$$\sum_{k=0}^{n} k.$$

Example 2

Rewrite using $j = n - k$:

$$\sum_{k=0}^{n} (2 + 3k).$$

Example 3

Use index reversal to evaluate

$$S = \sum_{k=0}^{n} (4 + 2k).$$

Combining Overlapping Sums

The lecture gives

$$\boxed{\sum_{k\in K} a_k + \sum_{k\in K'} a_k = \sum_{k\in K\cup K'} a_k + \sum_{k\in K\cap K'} a_k}$$

Main idea

If the two ranges overlap, the elements in the intersection have been counted twice.

Example 1

Simplify:

$$\sum_{k=1}^{3} a_k + \sum_{k=3}^{5} a_k.$$

Example 2

Simplify:

$$\sum_{k=1}^{5} a_k + \sum_{k=4}^{8} a_k.$$

Example 3

Rewrite as one main sum plus the repeated term:

$$\sum_{k=1}^{m} a_k + \sum_{k=m}^{n} a_k, \qquad 1 \le m \le n.$$

Multiple Sums

The general form is

$$\sum_{i \subset K_1,\ j \subset K_2} a_{i,j}$$

or, as a nested sum,

$$\sum_{i} \sum_{j} a_{i,j}.$$

What do the symbols mean?

- i, j — two indices;
- K_1, K_2 — sets of allowed index values;
- a_{ij} — a term depending on both indices.

How to use it

For every value of the outer index, evaluate the entire inner sum.

Example 1

Evaluate:

$$\sum_{i=1}^{2} \sum_{j=1}^{2} (i + j).$$

Example 2

Evaluate:

$$\sum_{i=1}^{3} \sum_{j=1}^{2} ij.$$

Example 3

Evaluate:

$$\sum_{i=1}^{3} \sum_{j=1}^{3} (i^2 + j).$$

Independent Indices

If the restrictions are independent, for example,

$$1 \le j \le n, \qquad 1 \le k \le m,$$

the order of summation can be changed:

$$\boxed{\sum_j \sum_k a_{jk} = \sum_k \sum_j a_{jk}.}$$

What does "independent" mean?

The allowed values of j do not depend on k, and the allowed values of k do not depend on j.

Example 1

Change the order of summation:

$$\sum_{j=1}^{2} \sum_{k=1}^{3} (j+k).$$

Example 2

Change the order and evaluate:

$$\sum_{j=1}^{3} \sum_{k=1}^{2} jk.$$

Example 3

Evaluate in two different orders:

$$\sum_{j=1}^{4} \sum_{k=1}^{3} (2j-k).$$

Product of Two Sums

If

$$a_{jk} = a_j b_k,$$

then

$$\boxed{\sum_{j\in J,k\in K} a_j b_k = \left(\sum_{j\in J} a_j\right)\left(\sum_{k\in K} b_k\right).}$$

How to use it

If the summand can be separated into one part depending only on j and another part depending only on k, the double sum can be written as the product of two ordinary sums.

Example 1

Evaluate:

$$\sum_{j=1}^{3}\sum_{k=1}^{2} jk.$$

Example 2

Simplify:

$$\sum_{j=1}^{n}\sum_{k=1}^{m} jk^2.$$

Example 3

Evaluate:

$$\sum_{j=1}^{4}\sum_{k=1}^{3}(2j)(k+1).$$

Dependent Indices

Consider

$$\boxed{\sum_{j=1}^{n}\sum_{k=j}^{n} a_{jk}.}$$

Here the allowed values of k depend on j.

The index region is

$$\boxed{1 \le j \le k \le n}.$$

When we change the order of summation,

$$\boxed{\sum_{j=1}^{n}\sum_{k=j}^{n} a_{jk} = \sum_{k=1}^{n}\sum_{j=1}^{k} a_{jk}.}$$

This is one of the main ideas of the **Multiple Sums** section.

How to use it

Do not simply exchange j and k. First determine the region described by the inequalities, and then describe the same region in the opposite order.

Example 1

Change the order of summation:

$$\sum_{j=1}^{4}\sum_{k=j}^{4} a_{jk}.$$

Example 2

Change the order and evaluate:

$$\sum_{j=1}^{3}\sum_{k=j}^{3} 1.$$

Example 3

Change the order and evaluate:

$$\sum_{j=1}^{4}\sum_{k=j}^{4} (j+k).$$

Triangular Double Sum

The lecture considers

$$S_U = \sum_{j=1}^{n} \sum_{k=j}^{n} a_j a_k.$$

The result is

$$\boxed{S_U = \frac{1}{2}\left[\left(\sum_{k=1}^{n} a_k\right)^2 + \sum_{k=1}^{n} a_k^2\right].}$$

What does this formula mean?

The condition

$$j \leq k$$

selects the upper triangular part of the matrix of products $a_j a_k$.

The formula allows us to calculate this triangular sum without adding every term individually.

Example 1

For

$$a_1 = 1, \qquad a_2 = 2, \qquad a_3 = 3,$$

calculate

$$\sum_{j=1}^{3} \sum_{k=j}^{3} a_j a_k.$$

Example 2

Evaluate

$$\sum_{j=1}^{4} \sum_{k=j}^{4} jk.$$

Example 3

Evaluate

$$\sum_{j=1}^{n} \sum_{k=j}^{n} jk$$

and express the answer in terms of n.